

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)
RVS NAGAR, CHITTOOR

III B.Tech-I Semester: Sessional Examination-I

Subject name: – Natural Language Processing and Models Subject code: 23ACM01 Time: 2 Hours

Branch :CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks 5 x 2 =10 M

1. Define lexical semantics and give an example.
2. How Pragmatics are used in NLP.
3. Define the concept of probability and its application in language modelling.
4. Give an example of Unigram Model.
5. Define perplexity in NLP.

Part B

Answer all the questions, each question carry equal marks 3 x 5 =15 M

- 6.(a) Describe the various types of lemmatization approaches.

OR

- 6.(b) Explain in detail about stopword removal process with example.

- 7.(a) Discuss about the Parts of Speech (POS) Tagging in NLP.

OR

- 7.(b) List and describe the various types of evaluation metrics.

- 8.(a) Demonstrate the FastText method with a python program.

OR

- 8.(b) Demonstrate the Naive Bayes theorem with a suitable example for text classification

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III B.Tech-I Semester: Sessional Examination-I

Subject name: – Operating Systems & System Programming

Subject code: 23ACA06

Time: 2 Hours

Branch :CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 =10 M

1. What is the difference between a process and a thread?
2. List the different process states in a process life cycle.
3. List any four CPU scheduling criteria used to evaluate scheduling algorithms.
4. What is a file? Mention any two basic file operations.
5. Define internal and external fragmentation with examples.

Part B

Answer all the questions, each question carry equal marks

3 x 5 =15 M

- 6.(a) Explain in details about Deadlock detection and Recovery with example.

OR

- 6.(b) Explain in details about the critical section problem.

- 7.(a) Describe about Peterson's solution with suitable example.

OR

- 7.(b) Describe how page tables are used and how logical addresses are translated to physical addresses.

- 8.(a) Describe about segmentation with a neat diagram.

OR

- 8.(b) Describe contiguous memory allocation with suitable example.

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III B.Tech-I Semester: Sessional Examination-I

Subject name: – Data Analysis with Python

Subject code: 23ACD11

Time: 2 Hours

Branch :CSM & CAI

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 =10 M

1. What is Anaconda Navigator?
2. Mention any two Python IDEs used for data analysis.
3. Define Pandas Series.
4. Mention any two challenges in data preprocessing.
5. Define normalization.

Part B

Answer all the questions, each question carry equal marks

3 x 5 =15 M

- 6.(a) Discuss the difference between NumPy Arrays, Pandas Series, and Data Frames.

OR

- 6.(b) Explain Pandas operations with examples.

- 7.(a) Describe how to set up Python environments using Jupyter, Anaconda, and VS Code

OR

- 7.(b) Compare Label Encoding and One-hot Encoding with examples.

- 8.(a) Explain normalization and standardization with examples.

OR

- 8.(b) Evaluate the role of preprocessing in data analysis.

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III B.Tech-I Semester: Sessional Examination-I

Subject name: – Introduction to Quantum Technologies and Applications Subject code: 23ACS27
Branch : CIVIL, EEE, ME, ECE, ECB, CSE, CSM, IT, CAI, **CSC & IOT, CSBS** Time: 2 Hours
Max Marks: 25

Part A

Answer all the questions, each question carry equal marks $5 \times 2 = 10$ M

1. Define superposition in quantum mechanics.
2. State the Heisenberg Uncertainty Principle in words.
3. Give one example illustrating wave-particle duality.
4. Define a qubit in simple terms.
5. Mention two differences between classical bits and quantum bits.

Part B

Answer all the questions, each question carry equal marks $3 \times 5 = 15$ M

6. (a) Explain the concept of superposition and its implications for quantum computing.

OR

6. (b) Discuss the phenomenon of entanglement and its role in quantum communication.
7. (a) State and explain the Uncertainty Principle, and give one experimental example demonstrating it.

OR

7. (b) Discuss the phenomenon of quantum decoherence and its effect on quantum systems.
8. (a) Explain the concepts of Hilbert spaces, quantum states, and operators in an abstract, intuitive way.

OR

8. (b) What is quantum coherence? Explain why it is essential for quantum computation.